Question

A binary compound M_xN_y crystal has the same lattice structure as NaCl crystal. If M is the vertex, then the coordinates of one N atom are $(1/2, 2/3, 2/3)$. Let the cell parameter be $a(pm)$, and the shortest distance between M and N be $d(pm)$.

Select the correct description below.

A. There are 24 N atoms in one unit cell of this crystal

B. The coordination number of M in this crystal is 12

C. In this crystal, M is situated in a coordination environment of a regular polyhedron.

D. d and a satisfy the relation $d = \sqrt{5}/6a$

E. The shortest distance between N and N in the crystal is $a/3$

F. In a unit cell with M as the vertex, the body diagonal passes through 4 atoms within the unit cell (excluding the vertices).

Answer

Correct Answer: D

Detailed Explanation

This is a face-centered cubic crystal.

$(1/2, 2/3, 2/3)$ undergoes C_3 axis and mirror operations to yield 12 points, namely $(1/2, \pm 2/3, \pm 2/3), (\pm 2/3, 1/2, \pm 2/3), (\pm 2/3, \pm 2/3, 1/2)$. Considering the three translation operations in the face-centered cubic lattice, a total of $12 \times (3+1) = 48$ points can be generated, i.e., $y = 48$. Therefore, option A is incorrect.

CHECKPOINT 3 PTS

There are 48 N atoms in a unit cell

In this crystal, N forms a truncated octahedron around M, which is not a regular polyhedron, with a coordination number of 24. Therefore, options B and C are incorrect.

CHECKPOINT 2 PTS

The coordination polyhedron of N around M is a truncated octahedron

CHECKPOINT 2 PTS

The coordination number of N around M is 24

The shortest distance between M and N in this crystal is the distance from $(0, 0, 0)$ to $(0, 1/6, 1/3)$, i.e., $d = \sqrt{5}/6a$, so option D is correct.

CHECKPOINT 2 PTS

$d = \sqrt{5}/6a$

The shortest distance between N and N in this crystal is the distance from $(0, 1/6, 1/3)$ to $(0, 1/3, 1/6)$, which is $\sqrt{2}/6a$, so option E is incorrect.

CHECKPOINT 2 PTS

The shortest N-N distance is $\sqrt{2}/6a$

In this crystal, apart from the vertices, no other atoms lie along the body diagonal, so option F is incorrect.

CHECKPOINT 1 PTS

No atoms lie on the body diagonal except at the vertices